Power Analysis Report Lab 10

Brady Adams, Jaiden Medina

Background:

The goal of this analysis is to use CubeMX and it's Power Consumption Calculator tool to accurately estimate the power consumption and expected battery life of the embedded system. The system worked on in the previous labs is an STM32L432 microcontroller and a TSL237 light-to-frequency sensor, powered by a 3.0 V CR2032 coin cell battery as described in the Lab description. The device is designed to periodically wake for 15 minutes, sample ambient light, and then return to a low-power sleep state.

For this lab, it is assumed that the system takes one light sample every 15 minutes during a 12-hour active period each day. Each sampling operation lasts approximately 15 seconds, during which the microcontroller is active, and the light sensor is enabled. When the system is not active, it remains in a low-power mode to minimize energy consumption.

For calculating the total current consumption, CubeMX only accounts for the microcontroller's power usage, so the current consumption of the TSL237 light sensor was included manually. According to the TSL237 datasheet, the sensor draws approximately 2 mA when active and negligible current when disabled. This sensor current was added during the 15-second active sampling period to calculate total system power consumption.

Power Mode Analysis:

Multiple combinations of running modes and low-power modes were evaluated to understand their impact on overall system power consumption and battery life. The running modes examined were Run and Low-Power Run, and the low-power modes examined were Stop0, Stop1, Stop2, and Standby.

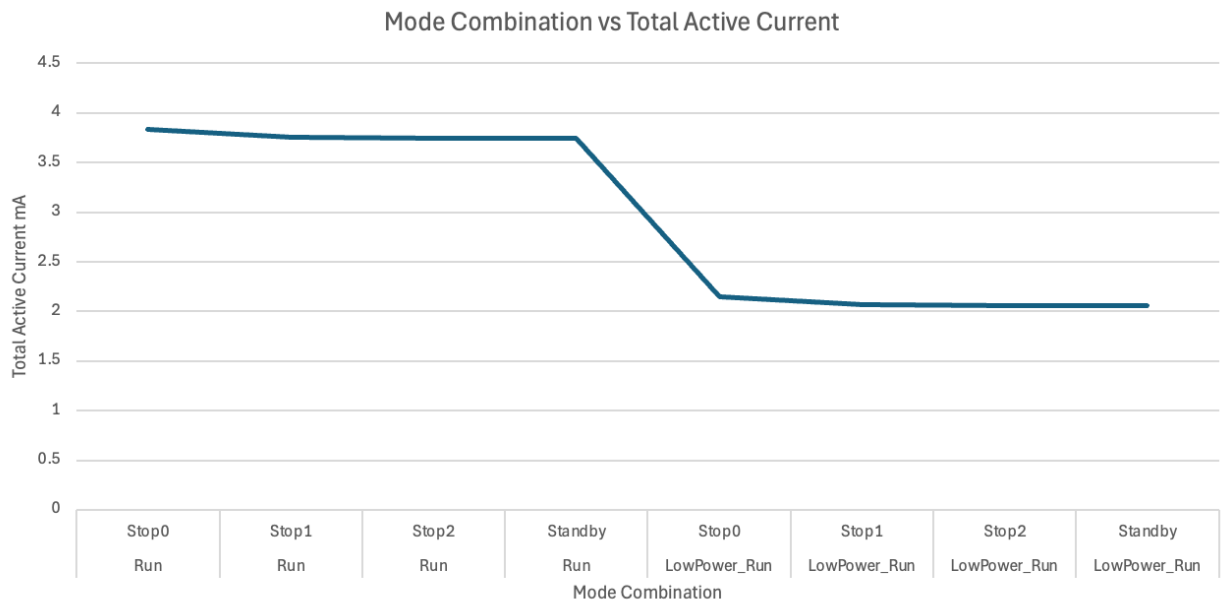

Figure 1 shows the total average current consumption for each mode combination, including both the MCU current reported by CubeMX and the manually added TSL237 sensor current.

From the data, configurations using the standard Run mode result in significantly higher total current consumption. For Run mode combined with Stop0, Stop1, or Stop2, the total average current is approximately 3.7–3.9 mA. This is primarily due to the higher baseline current of the microcontroller during active operation.

In contrast, using Low-Power Run significantly reduces the system's total current draw. When combined with Stop modes or Standby, the total average current drops to roughly 2.06–2.15 mA. This reduction demonstrates the use of the STM32L432's low-power operating modes in extending battery life.

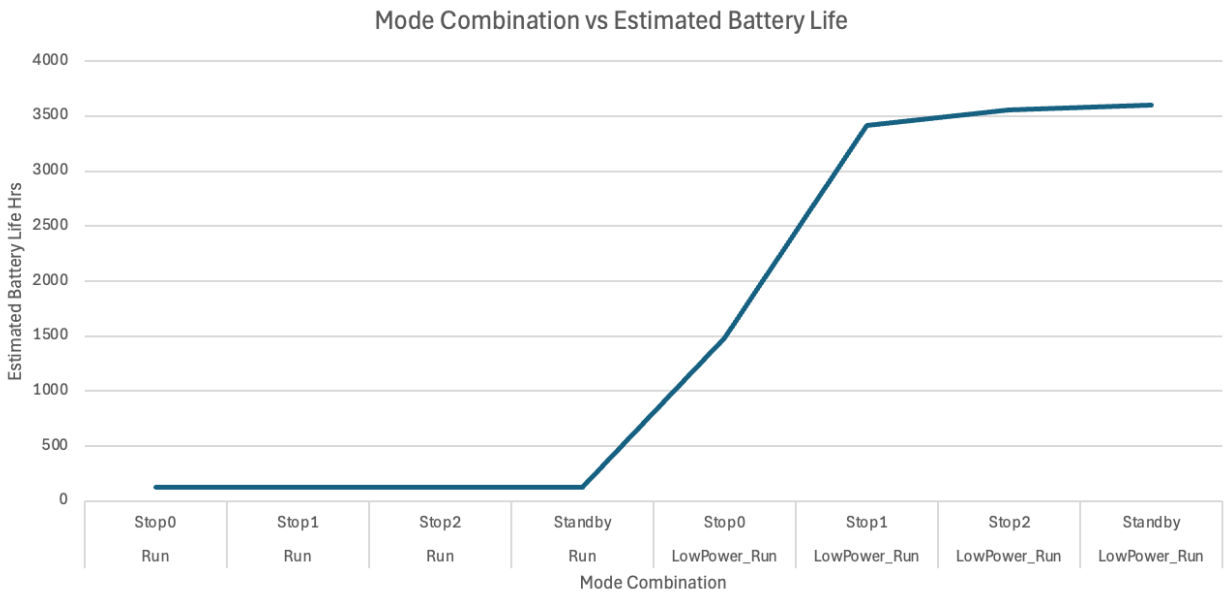

Figure 2 shows the estimated battery life for each mode combination based on the CubeMX calculations and total system current. The battery life is reported in hours and reflects the long-term energy impact of the selected power management strategy.

Systems operating in Run mode achieve relatively short battery lifetimes, roughly 120–130 hours. This indicates that continuously using the standard run configuration is unsuitable for a battery-powered sensor designed for long-term deployment.

By comparison, systems using Low-Power Run achieve dramatically longer battery life. Low-Power Run combined with Stop0, Stop1, or Stop2 yields battery life estimates exceeding 1,400 hours, with the longest lifetimes observed when combining Low-Power Run with Stop2 or Standby. These configurations exceed 3,500 hours, showing nearly an order-of-magnitude improvement over standard Run mode.

High Power Mode	Low Power Mode	MCU Average Current	TSL237 Sensor Current	Active Time	Total Active Current mA	Estimated Battery Life Hrs
Run	Stop0	1.83mA	2mA	15s	3.83	122
Run	Stop1	1.75mA	2mA	15s	3.75	128
Run	Stop2	1.74mA	2mA	15s	3.74	129
Run	Standby	1.74mA	2mA	15s	3.74	129
LowPower_Run	Stop0	0.15mA	2mA	15s	2.15	1488
LowPower_Run	Stop1	0.07mA	2mA	15s	2.07	3414
LowPower_Run	Stop2	0.06mA	2mA	15s	2.06	3558
LowPower_Run	Standby	0.06mA	2mA	15s	2.06	3603

Conclusion and Recommended Power Strategy:

Based on the results from the Power Consumption Calculator, the most effective power management strategy for this system is to operate the microcontroller in Low-Power Run mode during active sampling and transition to Stop2 or Standby mode during idle periods. This

approach minimizes average current consumption while still allowing timely wake-ups for periodic light measurements.

Overall, using Low-Power Run with Stop2 or Standby maximizes battery life while meeting the functional requirements of the system. This configuration enables long-term, energy-efficient operation and is the most appropriate choice for a battery-powered environmental light sensor.